features—like spin values about distinct axes—for which definite values are forbidden by the uncertainty principle.

But where could they have gone wrong? Well, remember that the Einstein, Podolsky, and Rosen argument hangs on one central assumption: if at a given moment you can determine a feature of an object by an experiment done on another, spatially distant object, then the first object must have had this feature all along. Their rationale for this assumption was simple and thoroughly reasonable. Your measurement was done over *here* while the first object was way over *there*. The two objects were spatially separate, and hence your measurement could not possibly have had any effect on the first object. More precisely, since nothing goes faster than the speed of light, if your measurement on one object were somehow to cause a change in the other—for example, to cause the other to take on an identical spinning motion about a chosen axis—there would have to be a delay before this could happen, a delay at least as long as the time it would take light to traverse the distance between the two objects. But in both our abstract reasoning and in the actual experiments, the two particles are examined by the detectors at the *same* time. Therefore, whatever we learn about the first particle by measuring the second must be a feature that the first particle possessed, completely independent of whether we happened to undertake the measurement at all. In short, the core of the Einstein, Podolsky, Rosen argument is that *an object over there does not care about what you do to another object over here.*

But as we just saw, this reasoning leads to the prediction that the detectors should find the same result more than half the time, a prediction that is refuted by the experimental results. We are forced to conclude that the assumption made by Einstein, Podolsky, and Rosen, no matter how reasonable it seems, cannot be how our quantum universe works. Thus, through this indirect but carefully considered reasoning, the experiments lead us to conclude that *an object over there does care about what you do to another object over here.*

Even though quantum mechanics shows that particles randomly acquire this or that property when measured, we learn that the randomness can be linked across space. Pairs of appropriately prepared particles— they're called *entangled* particles—don't acquire their